

PRODUCER ANALYSIS + GENERATIVE MUSIC SPECIFICATION

HOUSE MUSIC MIDI GENERATOR BRAIN

A practical system for chords, melody, bass, drums, groove, and arrangement - built from broad traits associated with Prospa, Chris Stussy, KETTAMA, Avicii, and David Guetta.

Purpose: give a MIDI generator enough musical context to create original house ideas with controlled stylistic direction, while avoiding note-for-note reproduction of released records.

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Prepared as a model-facing knowledge document



Scope, evidence, and the originality rule

Important framing

This is an independent educational analysis. It is not endorsed by the named artists, and it does not claim access to their private sessions, stems, MIDI files, presets, or unreleased methods. Public statements are identified as documented context. Musical rules labeled as analysis are inferences from broad, audible genre traits.

The named artists do not share one formula. Prospa's rave-charged house language, Chris Stussy's low-slung minimal/deep-tech groove, KETTAMA's high-impact loop pressure, Avicii's melodic progressive writing, and David Guetta's pop-facing EDM construction occupy different regions of the house family. A useful generator should therefore treat them as separate parameter presets, not as interchangeable tags.

This document describes composition at the level of rhythm, harmony, contour, density, and arrangement. It deliberately avoids recreating identifiable hooks. A generator may learn that a style uses a four-bar diatonic loop, a syncopated chord stab, or a rolling offbeat bass; it should not store or emit the exact pitch-and-rhythm sequence of a copyrighted melody.

Three evidence labels used throughout

LABEL	MEANING	HOW TO USE IT
DOCUMENTED	A workflow point stated in a public interview, official artist page, or DAW manual.	Safe as contextual evidence; still not a complete description of an artist's process.
ANALYSIS	A generalized listening-based inference about harmony, groove, density, or form.	Use as a weighted prior, never as a rigid truth about every track.
GENERATOR RULE	An original, implementation-focused constraint derived from theory and analysis.	Use directly in code, prompts, or a symbolic music model.

The originality rule

The generator should optimize for family resemblance at the level of musical grammar, not identity at the level of a hook. Enforce at least four independent changes whenever a seed resembles a reference: transpose the key, alter the harmonic rhythm, change the melodic contour, and mutate the onset pattern. In addition, reject any melody whose interval-and-rhythm fingerprint is too similar to a protected or curated reference set.

What this PDF can be used for

- Retrieval-augmented generation: chunk the PDF by heading and retrieve the relevant archetype before generation.
- Prompt context: paste a parameter card, progression bank, and drum grid into an LLM or symbolic generator.
- Rule-based MIDI: implement the pseudocode and weighted constraints in a piano-roll or MIDI scripting system.
- Dataset labeling: convert each archetype into style, harmony, melody, groove, and arrangement labels.
- Human writing: use the templates as starting points, then replace at least one major musical assumption.

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PART I

The common MIDI and music model

Before an artist archetype can matter, the generator needs a stable representation of time, pitch, harmony, velocity, probability, and form. These chapters define that shared substrate.

1. MIDI is performance data, not sound

A MIDI generator does not create audio. It creates instructions that a synthesizer, sampler, or drum rack interprets. The musical intelligence therefore lives in the relationships among note number, onset, duration, velocity, channel, and optional expressive controls. Sound design can transform the emotional result, but a weak symbolic pattern remains weak.

Store note timing in ticks relative to a quarter note, commonly called PPQ. A resolution of 480 PPQ is sufficient for most house grooves; 960 PPQ gives finer microtiming. At 480 PPQ, a sixteenth note is 120 ticks. A swing delay of 18 ticks equals 15 percent of a sixteenth-note cell. Keep the canonical pattern grid separate from expressive offsets so patterns can be quantized, compared, and mutated without losing the intended groove.

Minimum note-event schema

```
{
  "track_role": "chord|melody|bass|kick|clap|hat|perc",
  "pitch": 60,
  "start_tick": 0,
  "duration_tick": 120,
  "velocity": 96,
  "probability": 1.0,
  "micro_offset_tick": 0,
  "articulation": "normal|short|legato|ghost|accent"
}
```

Separate intention from rendering

LAYER	REPRESENTATION	EXAMPLE
Musical intention	Scale degree, chord function, onset class, role	Degree 3 over vi; late offbeat stab; pickup note
MIDI realization	Pitch, tick, duration, velocity, channel	E4, tick 840, 90 ticks, velocity 84
Sound realization	Instrument, envelope, filter, effects, mix	Short organ stab, high-pass, plate reverb

Recommended internal time model

- **Bar:** four quarter-note beats in 4/4.
- **Step:** one sixteenth note; 16 steps per bar.
- **Phrase:** usually four or eight bars.
- **Section:** 8, 16, or 32 bars, with exceptions for fills and pickups.
- **Grid time:** exact quantized location.
- **Performance time:** grid time plus swing and microtiming offset.

Velocity is structural

Velocity should encode accent hierarchy, not merely random loudness. Strong kick and clap anchors usually occupy a narrow high range. Closed hats can alternate strong and weak values. Ghost percussion should be clearly lower. Chord voicings benefit from small internal differences, but moving every note independently can blur what should feel like one gesture.

The Ableton manual documents per-note velocity and chance editing, while groove extraction can carry timing and velocity information. Those features reflect a general principle for any generator: probability, velocity, and timing are separate axes. A note can be likely but quiet, rare but accented, or perfectly quantized but dynamically human.

Baseline velocity bands

ROLE	NORMAL	GHOST	ACCENT	VARIATION POLICY
Kick	105-120	72-88	118-127	Narrow; avoid randomizing the four floor anchors heavily
Clap / snare	96-114	58-76	112-124	Main hits stable; pre-claps and ghosts can vary
Closed hat	62-94	40-62	90-108	Alternate strong/weak and add small random changes
Open hat	82-108	60-78	102-118	Protect offbeat accents
Percussion	52-100	34-58	92-112	Largest range; use probability with role awareness
Chord	70-105	55-72	100-116	Keep notes within one chord mostly coherent
Lead	76-112	60-74	104-120	Accent phrase peaks and important motif notes
Bass	84-116	64-80	110-124	Stable roots; lower passing-note velocity

2. Harmony engine: function first, notes second

Generate harmony in scale degrees or Roman numerals before choosing absolute MIDI pitches. This makes transposition trivial and allows one progression to be voiced differently for each archetype. The same functional loop can become bright festival piano, muted deep-house sevenths, rave organ stabs, or a sparse two-note shell.

For major keys, the diatonic triads are I major, ii minor, iii minor, IV major, V major, vi minor, and vii diminished. For natural minor, they are i minor, ii diminished, III major, iv minor, v minor, VI major, and VII major. House music frequently treats the relative major and minor as a shared palette, and many loops do not create a strong classical cadence. That ambiguity is useful.

Chord-object schema

```
{
  "root_degree": 6,
  "quality": "minor",
  "extensions": [
    7,
    9
  ],
  "inversion": 1,
  "bass_degree_override": null,
  "start_step": 0,
  "length_steps": 4,
  "voicing_span_semitones": 19,
  "top_note_target_degree": 3
}
```

A practical voicing algorithm

- Build the chord from its scale-degree root and quality.
- Add extensions according to archetype probability: sevenths and ninths are common in deeper house; plain triads often read more directly in pop-progressive writing.
- Choose an inversion that minimizes total voice movement from the previous chord.
- Constrain the chord register. Typical MIDI center: 55-72 for piano/organ body, 64-84 for bright synth stacks.
- If bass owns the root, omit or raise the chord's lowest root to reduce mud.
- Treat the top note as a melodic voice. Prefer stepwise movement or a repeated common tone.
- Apply rhythmic articulation only after the pitch voicing is accepted.

Voice-leading cost

A simple cost function can compare candidate inversions. Sort chord notes by pitch, match voices, then sum absolute semitone movement. Add penalties for crossings, register violations, duplicated low thirds, and leaps greater than seven semitones. Subtract a small reward when a common tone stays in place.

```
cost = sum(abs(next_voice[i] - prev_voice[i]))
+ 5 * voice_crossings
+ 3 * register_violations
+ 2 * count(leap > 7 semitones)
```

- 1.5 * stationary_common_tones

Harmonic rhythm choices

PATTERN	STEPS PER CHORD	CHARACTER	BEST FIT
Whole bar	16	Broad, songlike, easy to sing	Avicii / Guetta archetypes
Half bar	8	Forward motion without clutter	All five
Quarter bar	4	Driving and declarative	Rave and festival passages
Syncopated stabs	1-3 with rests	Groove becomes part of harmony	Prospa / Stussy / KETTAMA
Pedal harmony	Chord changes above held bass or tone	Tension and continuity	Deep/minimal and melodic builds

Borrowed-color rules

Borrowed chords should be used as controlled color, not random wrong notes. In a major key, iv minor can add bittersweet pull; bVII can create rock or rave openness; a secondary dominant can intensify arrival. In natural minor, a major V introduces the leading tone and a stronger cadence. For generator safety, limit chromatic chord events to zero or one per four-bar loop unless the preset explicitly requests modal or rave tension.

When a borrowed chord appears, the melody must either accommodate its altered pitch or avoid the conflicting scale degree. Harmony and melody should be generated jointly enough to prevent a natural scale note from colliding with the chord's chromatic tone.

3. Melody engine: motif, contour, and chord awareness

A convincing lead is usually not a random walk through a scale. It is a short motif with an identity, repeated with controlled changes. Store the motif as intervals and onset gaps so it can be transposed, reharmonized, and rhythmically mutated. The generator should know which notes are structural chord tones and which are passing, neighbor, anticipation, or suspension tones.

A useful four-bar melody often behaves like a sentence: establish an idea, repeat or answer it, expand or lift it, then close or leave a pickup. That behavior can be expressed without copying any specific melody.

Melody event annotations

```
{
  "scale_degree": 3,
  "octave": 5,
  "onset_step": 6,
  "duration_steps": 1,
  "function": "chord_tone|passing|neighbor|anticipation|suspension",
  "motif_index": 2,
  "phrase_role": "statement|answer|lift|cadence|pickup"
}
```

Strong-beat note policy

LOCATION	DEFAULT PITCH POLICY	EXCEPTION
Beat 1	Chord tone, often root/third/fifth or a held anticipation	A suspension may resolve within one or two eighths
Beat 2 / 4	Chord tone or stable scale tone	Groove-led archetypes can accent non-chord tones briefly
Offbeat eighth	Passing, neighbor, chord extension, or anticipation	Can be the main hook in syncopated house
Sixteenth pickup	Scale step toward target	Chromatic approach only if explicitly enabled
Phrase ending	Root, third, fifth, or unresolved second/fifth	Open-ended loops may avoid tonic resolution

Contour grammar

- **Statement:** small range, clear rhythm, 2-5 distinct pitches.
- **Answer:** keep rhythm, change final interval or starting scale degree.
- **Lift:** raise register or peak by 2-5 scale steps; increase density slightly.
- **Cadence:** reduce density and lengthen the last note, or insert a pickup into the loop.
- **Range guard:** default total range 7-12 semitones for vocal-like hooks; 12-19 for energetic instrumental leads.
- **Leap recovery:** after a leap greater than a perfect fourth, prefer stepwise motion in the opposite direction.

Motif mutation operators

OPERATOR	DESCRIPTION	SAFE RANGE
Transpose degree	Move the full motif diatonically, then fit chord tones	+/- 1 to 3 scale degrees
Rhythmic displacement	Shift selected onsets while preserving order	+/- one sixteenth or one eighth
Tail rewrite	Keep first 60-75% and regenerate ending	One or two notes changed
Compression	Replace a leap with stepwise motion	Reduce interval by 2-5 semitones
Expansion	Increase one important interval	Add 2-4 semitones, obey range
Fragmentation	Use only the first cell, separated by rests	1-3 repetitions per bar
Augmentation	Double durations	Best in breakdown or vocal space
Diminution	Halve durations	Best in build or high-energy answer

Catchiness without copying

Catchiness can be optimized with compactness, repetition, rhythmic identity, and a controlled peak. It does not require similarity to an existing hook. Score a melody positively for motif reuse, singable range, stable strong beats, a memorable onset pattern, and clear phrase direction. Penalize featureless scalar runs, too many equal note values, excessive large leaps, and exact repetition without a phrase-level change.

Create an interval fingerprint as signed semitone intervals and a rhythm fingerprint as normalized onset gaps. Compare both with a reference database. If pitch intervals and rhythm both cross a similarity threshold, regenerate the phrase even after transposition, because transposition alone does not create a new melody.

4. Drum and groove engine

House drums are a hierarchy of anchors, propulsion, and conversation. The kick establishes the floor. Clap or snare marks the backbeat. Closed and open hats define subdivisions and swing. Percussion adds syncopation and phrase detail. A generator should create these roles in that order, because randomly filling a 16-step grid tends to produce density without groove.

Represent every lane as an onset mask plus velocity, probability, and microtiming arrays. Use deterministic anchors and probabilistic decorations. A fill should be generated as a transition event tied to section boundaries, not as permanent complexity.

Canonical four-on-the-floor grid

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Open hat			X				X				X				X	
Shaker		X		X		X		X		X		X		X		X
Perc ghost				o				o				o			o	

Legend: filled dot = primary hit; open dot = ghost or probability-controlled hit. Grid cells are sixteenth notes.

Swing model

Apply swing to the second subdivision in each pair, not to all notes indiscriminately. For a sixteenth-note grid, steps 2, 4, 6, 8, 10, 12, 14, and 16 may receive a positive delay. A normalized swing value of 0.0 is straight. A practical house range is roughly 0.05-0.28 of one sixteenth cell, depending on archetype. Preserve kick anchors unless the preset explicitly permits a loose kick.

```
performed_tick = grid_tick
+ swing_amount * step_ticks if step_index is the swung subdivision
+ lane_micro_offset
+ random_jitter
```

Lane-specific timing

LANE	TYPICAL OFFSET	WHY
Kick	0 to +4 ms	Stable anchor; excessive jitter weakens the floor
Clap / snare	+2 to +14 ms	A slightly late backbeat can feel relaxed and wide
Closed hat	-4 to +10 ms plus swing	Pushes or relaxes subdivision feel
Open hat	0 to +8 ms	Protects the offbeat while avoiding machine stiffness
Shaker	-6 to +12 ms plus swing	Can carry most of the human movement
Percussion	-12 to +18 ms	Conversation around the anchors

Probability policy

- Kick anchors: 0.98-1.00 probability.
- Main claps: 0.99-1.00.
- Closed-hat backbone: 0.85-1.00; decorative hats: 0.25-0.75.
- Ghost percussion: 0.20-0.60, with no more than two simultaneous optional events on one step.
- Fills: conditional on section transition, typically 0.35-0.80 depending on energy.
- Never make all optional lanes decide independently. Use a density budget per bar.

Groove-coherence test

Mute every lane except kick, clap, and one hat. If the pattern does not communicate a stable pulse and subdivision feel, adding more percussion will not fix it. In code, measure onset balance by beat, offbeat coverage, syncopation count, and maximum simultaneous hits. Reject bars that exceed the preset's density budget or leave an unintended empty region.

5. Bass, chords, and kick as one system

Bass generation should happen after the harmonic skeleton and kick pattern are known. The bass may reinforce roots, answer the chord rhythm, or act as a counter-melody. Its note lengths must leave room for kick transients and avoid uncontrolled overlap across chord changes.

For a root-led festival pattern, choose mostly roots and fifths with clean offbeat timing. For rolling minimal house, derive a two-bar rhythmic cell with rests, repeats, and occasional approach notes. For rave pressure, use short repeated roots, octave jumps, or sampled bass stabs. In every case, store pitch choice and rhythmic choice separately so either can mutate without destroying the other.

Bass-note selection weights

CONTEXT	ROOT	FIFTH	THIRD	OCTAVE	PASSING / APPROACH
Pop-progressive	0.55	0.18	0.08	0.15	0.04
Minimal / deep tech	0.40	0.15	0.10	0.15	0.20
Rave / high energy	0.48	0.12	0.05	0.25	0.10

Collision rules

- If a bass onset lands exactly with a kick, shorten bass attack or delay by a few milliseconds only if the style permits.
- Avoid holding a bass third below a dense chord voicing in the same octave.
- At chord change, either update bass immediately, anticipate by one eighth, or deliberately sustain as a pedal. Do not drift accidentally.
- Keep most sub-bass notes between MIDI 28 and 48; exact range depends on sound and key.
- Use note length as groove: short bass creates space, while legato bass creates pressure and must be managed carefully.

Sidechain-aware MIDI

Even when audio sidechain compression will be applied later, the symbolic generator should create space. Include a per-event ducking hint or ensure bass and chord envelopes have release times compatible with the kick rate. MIDI cannot perform compression by itself, but it can avoid forcing the mix engine to solve unnecessary rhythmic conflicts.

PART II

Five artist-associated generator archetypes

Each profile is a probabilistic preset. Values are recommended generation ranges, not claims that every record by the artist uses them.

PROFILE 1

PROSPA

PROSPA: Rave-house lift, breakbeat memory, and piano-stab urgency

Documented context
Public profiles and interviews establish Prospa as a UK electronic duo whose work crosses house, rave, and breakbeat lineages. The exact compositional details below are analysis rather than a claim about a fixed private workflow.

Generator identity

PARAMETER	DEFAULT / RANGE
Tempo	124-134 BPM; default 128
Core meter	4/4
Harmonic loop	1-4 bars; often rhythmically articulated
Chord density	Medium to high in peaks; sparse in tension passages
Melody density	Motif-led, medium; short rave signals over long songlike lines
Swing / microtiming	0.06-0.20; straight anchors with shuffled or broken top layers
Kick policy	Four-on-floor or hybrid; breaks may interrupt or answer
Primary energy lever	Piano/organ stabs, rave motifs, breaks, octave lift

Listening-based analysis

- Treat the harmonic layer as a rhythmic instrument. A chord may be only one or two sixteenths long, with reverb or delay supplying the apparent sustain.
- Use bright major or suspended color against more forceful drums. Alternating uplift and pressure is more characteristic than constant harmonic complexity.
- Allow a breakbeat memory layer: shuffled snare ghosts, syncopated kicks, or chopped percussion can appear while the main house floor remains understandable.
- Motifs should feel like rave signals: compact pitch sets, octave reinforcement, repetitions, and strategically placed pickups.

Harmonic engine

ID	ROMAN NUMERALS	EXAMPLE IN KEY	GENERATOR USE
P1	i - VI - III - VII	A minor: Am - F - C - G	Uplifting minor loop; use piano stabs or wide synth
P2	I - V - vi - IV	E major: E - B - C#m - A	Bright hands-up loop; shorten chords for rave articulation
P3	vi - IV - I - V	C major: Am - F - C - G	Emotional but direct; move top note stepwise
P4	i - bVII - bVI - bVII	F minor: Fm - Eb - Db - Eb	Rave tension; strong bass roots and sampled chord color
P5	I(sus2) - IV - vi7 - V	D major: Dsus2 - G - Bm7 - A	Airy lift; extensions only in upper voices

VOICING RULE

Use compact midrange triads or sevenths, then double the top note one octave higher. Preserve one common tone when possible. For piano, velocities can fan from about 92 in the lower note to 104 in the top accent. Chord lengths of 1-4 sixteenth steps create urgency; longer breakdown chords can last a half or full bar.

Melody engine

- Start from a 2-4 note cell with one distinctive rhythm.
- Favor scale degrees 1, 2, 3, and 5; reserve 6 or 7 for lift or tension.
- Repeat the cell at a new chord tone, then change its tail.
- Use octave jumps as punctuation, not on every note.
- Place a one-eighth or one-sixteenth pickup before a downbeat arrival.
- In a breakbeat passage, sync the lead to snare or chopped-vocal accents rather than every kick.

One-bar drum skeleton

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Break snare ghost				o			o					o			o	
Closed hat			X				X				X				X	
Rave ride	o		o		o		o		o		o		o		o	
Perc		o				X		o				o		X		

Treat this grid as a skeleton. Primary hits are deterministic; open-dot positions are ghost or probability-controlled. Vary velocity and microtiming within the profile's groove range.

Bass engine

Default to short root or octave notes on offbeats, then add one anticipatory note into bars 2 or 4. When breakbeat energy rises, allow bass to mirror the broken kick for one bar. Keep the root obvious even when the rhythm becomes busy.

Arrangement state sequence

STATE	LENGTH	MIDI ACTION
DJ intro	16 bars	Kick, filtered loop, sparse break fragment
Pressure A	16 bars	Bass and main groove; one chord signal introduced
Lift	8 bars	Longer chords, riser, reduced kick or filtered floor
Peak A	16-32 bars	Full stabs, octave motif, ride or breaks
Switch	8 bars	Breakbeat-led contrast or bass dropout
Peak B / outro	16-32 bars	Return with variation, then simplify for mixing

Parameter card

```
{
  "archetype": "rave_house_lift",
  "tempo": {
    "min": 124,
    "max": 134,
    "default": 128
  },
  "key_mode_weights": {
    "minor": 0.58,
    "major": 0.32,
    "modal": 0.1
  },
  "chord_articulation": {
    "stab": 0.62,
    "sustain": 0.23,
    "rhythmic_gate": 0.15
  },
  "breakbeat_overlay_probability": 0.42,
  "lead_octave_jump_probability": 0.18,
  "swing_range": [
    0.06,
    0.2
  ],
  "section_phraseBars": [
    8,
    16,
    32
  ]
}
```

Generation recipe

- Choose a tempo inside 124-134 BPM and select a key from the weighted mode distribution.
- Select one progression template, transpose it, and generate three inversion candidates. Keep the lowest voice-leading-cost version.
- Generate a role-correct bass rhythm before decorative percussion. Reject collisions that weaken the kick.
- Create a short motif from scale degrees, then fit strong onsets to chord tones. Produce statement, answer, lift, and cadence variants.
- Apply the drum skeleton, optional-event budget, lane velocities, swing, and microtiming.
- Map the phrase into the arrangement state sequence. Every 4-8 bars, alter only one or two attributes.
- Run originality, range, density, collision, and phrase-coherence tests before export.

Failure modes

- Using only festival EDM supersaw language and omitting rave/break rhythmic memory.
- Making every chord a full-bar sustain, which removes percussive urgency.
- Adding breakbeats so densely that the house anchor disappears.
- Copying a recognizable piano rhythm instead of generating a new onset mask.

Originality mutations for this archetype

- Swap four-on-floor and broken-kick emphasis for one phrase without changing the harmony.
- Move the stab rhythm by one eighth and invert the chord top voice.
- Replace an octave lead jump with a fifth followed by stepwise recovery.
- Change the final bar from major lift to suspended openness.

PROFILE 2

CHRIS STUSSY

CHRIS STUSSY: Minimal/deep-tech pocket, rolling bass, and restrained harmonic color

Documented context

In public interviews, Chris Stussy has discussed using FL Studio and drum machines, and described disciplined studio practice. The detailed harmonic and groove rules here are generalized analysis of minimal/deep-tech conventions associated with his lane.

Generator identity

PARAMETER	DEFAULT / RANGE
Tempo	124-130 BPM; default 127
Core meter	4/4
Harmonic loop	2-8 bars; two-bar groove cells are central
Chord density	Low; one shell or stab can define the loop
Melody density	Low to medium; fragments and call-response
Swing / microtiming	0.14-0.30; project-wide groove with lane-specific offsets
Kick policy	Stable four-on-floor
Primary energy lever	Bass rhythm, shuffle, subtractive arrangement, micro-variation

Listening-based analysis

- The bassline is often the main melodic engine. Harmony can be a shell voicing, filtered sample, or short jazz-colored stab.
- Swing is not decoration applied at the end. It affects hats, percussion, bass, and chord placement as one pocket.
- Repetition should be long enough for the groove to hypnotize, while one or two small events change every 4-8 bars.
- Silence is part of the pattern. Avoid filling every sixteenth or changing harmony merely to create motion.

Harmonic engine

ID	ROMAN NUMERALS	EXAMPLE IN KEY	GENERATOR USE
S1	i7 - iv7	G minor: Gm7 - Cm7	Two-chord pocket; invert upper voices
S2	ii7 - V7	F major: Gm7 - C7	Unresolved jazz loop; bass need not play every root
S3	i9 - bVIIImaj7	D minor: Dm9 - Cmaj7	Warm color with stepwise top voice
S4	vi7 - ii7 - V7	C major: Am7 - Dm7 - G7	Longer four- or eight-bar phrase
S5	I6/9 pedal	Eb major: Eb6/9 over repeating Eb/Bb bass	Static harmony; movement comes from rhythm and filtering

VOICING RULE

Prefer 2-4 notes. Shell voicings can use root plus seventh, or third plus seventh if bass supplies the root. Add ninths selectively. Place chords around MIDI 55-76 and avoid thick low triads. Reuse common tones, and shift one inner voice by a semitone or whole step to create sophisticated motion without a new root.

Melody engine

- Generate a one-bar fragment with 1-4 notes, then leave space.
- Use chord extensions such as 7, 9, or 6 as color notes.
- Repeat rhythm more often than pitch; answer in a new register every 2-4 bars.
- Allow one chromatic approach note into a stable target when harmony supports it.
- Avoid a large anthem-like peak; keep range usually within 5-10 semitones.
- Treat sampled vocal or synth blips as melody roles even if pitch is ambiguous.

One-bar drum skeleton

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Shaker		X		X		X		X		X		X		X		X
Rim / wood				o			X					o			X	
Perc		o				X		o		o				X		

Treat this grid as a skeleton. Primary hits are deterministic; open-dot positions are ghost or probability-controlled. Vary velocity and microtiming within the profile's groove range.

Bass engine

Build a two-bar cell with 5-9 notes total, mixing roots, fifths, octave repeats, and one approach tone. Use short-to-medium durations and protect rests after important kicks. Repeat the exact rhythm for several phrases, then mutate one onset or final pitch.

Arrangement state sequence

STATE	LENGTH	MIDI ACTION
DJ intro	16-32 bars	Kick, hats, filtered texture; bass teased late
Groove A	32 bars	Full bass cell; minimal stab; small 8-bar variations
Break	8-16 bars	Remove kick or sub; expose chord/vocal fragment
Groove B	32 bars	Return with one new perc lane or bass tail
Reduction	16 bars	Subtract melodic color; focus on drums and bass
DJ outro	16-32 bars	Remove bass or signature fragment gradually

Parameter card

```
{
  "archetype": "minimal_deep_tech_pocket",
  "tempo": {
    "min": 124,
    "max": 130,
    "default": 127
  },
  "key_mode_weights": {
    "minor": 0.64,
    "major": 0.18,
    "dorian": 0.18
  },
  "chord_extension_probability": {
    "7": 0.72,
    "9": 0.42,
    "6": 0.22
  },
  "bass_notes_per_2_bars": [
    5,
    9
  ],
  "swing_range": [
    0.14,
    0.3
  ],
  "optional_drum_density_budget": 7,
  "mutation_period_bars": [
    4,
    8,
    16
  ]
}
```

Generation recipe

- Choose a tempo inside 124-130 BPM and select a key from the weighted mode distribution.
- Select one progression template, transpose it, and generate three inversion candidates. Keep the lowest voice-leading-cost version.
- Generate a role-correct bass rhythm before decorative percussion. Reject collisions that weaken the kick.
- Create a short motif from scale degrees, then fit strong onsets to chord tones. Produce statement, answer, lift, and cadence variants.
- Apply the drum skeleton, optional-event budget, lane velocities, swing, and microtiming.
- Map the phrase into the arrangement state sequence. Every 4-8 bars, alter only one or two attributes.
- Run originality, range, density, collision, and phrase-coherence tests before export.

Failure modes

- Writing four large chords per bar and a busy lead above them.
- Randomizing every lane independently, destroying the shared pocket.
- Changing the bassline every bar instead of letting repetition work.
- Confusing minimal with empty: the remaining events must have strong placement and tone.

Originality mutations for this archetype

- Keep bass rhythm but rotate pitches through a new chord shell.
- Remove one percussion event and add a lower-velocity answer two steps later.
- Hold harmony static while moving only an inner voice.
- Delay the chord stab by one swung sixteenth and shorten its duration.

PROFILE 3

KETTAMA

KETTAMA: High-impact loop pressure, hard drums, and towering chord signals

Documented context
KETTAMA has publicly described an intuitive FL Studio process, a sample-heavy approach, and an appetite for large chords, basslines, and hard drums. The concrete MIDI constraints below are generalized generator rules, not transcriptions.

Generator identity

PARAMETER	DEFAULT / RANGE
Tempo	130-145 BPM; default 136
Core meter	4/4
Harmonic loop	1-4 bars; aggressive loop identity
Chord density	Low to medium; large gesture, simple content
Melody density	Low to medium; hooks from samples, stabs, or short riffs
Swing / microtiming	0.04-0.18; forward and forceful
Kick policy	Hard four-on-floor, hybrid, or rave/techno drive
Primary energy lever	Drum impact, sample repetition, chord scale, tension drops

Listening-based analysis

- Start with a loop that already feels forceful before adding arrangement. A strong kick, bass shape, and one hook can carry the identity.
- Harmonic sophistication matters less than scale, register, and rhythm. One huge chord stab can outperform a detailed progression.
- Use repetition to create pressure, then create contrast by muting the kick, filtering the loop, or exposing a vocal/sample fragment.
- At faster tempo, shorten notes and simplify syncopation enough to preserve impact.

Harmonic engine

ID	ROMAN NUMERALS	EXAMPLE IN KEY	GENERATOR USE
K1	i - bVII - bVI - bVII	E minor: Em - D - C - D	Rave loop with direct descending gravity
K2	i pedal + bII stab	F minor: Fm with occasional Gb/F	Dark tension; use bII as rare accent
K3	i - VI	C minor: Cm - Ab	Two-chord pressure; octave bass and huge top voice
K4	I - bVII	A major/mixolydian: A - G	Open, blunt, sample-friendly
K5	vi - IV - V	D major: Bm - G - A	Three-chord lift for a larger melodic section

VOICING RULE

Use simple triads, fifth stacks, or sampled chord shapes with broad register. Double root or fifth in octaves and keep the midrange uncluttered. Stabs can be 1-2 sixteenths at high velocity; breakdown chords may extend for a bar. A detuned or sampled sound can provide complexity that the MIDI does not need to spell.

Melody engine

- Use 2-5 distinct pitches in a one- or two-bar hook.
- Favor minor pentatonic, natural minor, Dorian, or Mixolydian sets depending on mood.
- Repeat one onset pattern insistently; change register or last note after 2-4 bars.
- Allow octave displacement and repeated notes more than winding scalar lines.
- Keep pickups short at high BPM.
- If a sample is the hook, generate supporting MIDI that leaves its spectral and rhythmic space open.

One-bar drum skeleton

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Kick	X				X				X				X				
Clap / snare					X								X				
Closed hat			X		X		X				X		X		X		
Open hat			X				X				X				X		
Ride	X		X		X		X		X		X		X		X		
Perc / fill				o			X					o		X	X	X	

Treat this grid as a skeleton. Primary hits are deterministic; open-dot positions are ghost or probability-controlled. Vary velocity and microtiming within the profile's groove range.

Bass engine

Use short repeated roots, octave alternation, or a one-bar sampled-bass rhythm. Total density can be 4-10 notes per bar, but the sound envelope may make each note feel larger. Avoid decorative passing tones that weaken the blunt harmonic center.

Arrangement state sequence

STATE	LENGTH	MIDI ACTION
Impact intro	8-16 bars	Immediate loop identity; filtered or reduced low end
Drive A	16-32 bars	Full kick and bass; hook repeated
Tension cut	4-8 bars	Kick mute, chord tail, sample or vocal focus
Drive B	16-32 bars	Ride, octave lift, or denser hats
Major break	8-16 bars	Large chord or melodic contrast
Final pressure	16-32 bars	Full return with one variation and hard ending/outro

Parameter card

```
{
  "archetype": "high_impact_loop_pressure",
  "tempo": {
    "min": 130,
    "max": 145,
    "default": 136
  },
  "key_mode_weights": {
    "minor": 0.66,
    "major": 0.12,
    "dorian": 0.1,
    "mixolydian": 0.12
  },
  "hook_distinct_pitch_range": [
    2,
    5
  ],
  "chord_stab_duration_steps": [
    1,
    2
  ],
  "kick_anchor_probability": 1.0,
  "ride_entry_energy_threshold": 0.72,
  "tension_cut_bars": [
    4,
    8
  ]
}
```

Generation recipe

- Choose a tempo inside 130-145 BPM and select a key from the weighted mode distribution.
- Select one progression template, transpose it, and generate three inversion candidates. Keep the lowest voice-leading-cost version.
- Generate a role-correct bass rhythm before decorative percussion. Reject collisions that weaken the kick.
- Create a short motif from scale degrees, then fit strong onsets to chord tones. Produce statement, answer, lift, and cadence variants.
- Apply the drum skeleton, optional-event budget, lane velocities, swing, and microtiming.
- Map the phrase into the arrangement state sequence. Every 4-8 bars, alter only one or two attributes.
- Run originality, range, density, collision, and phrase-coherence tests before export.

Failure modes

- Adding complex jazz voicings where scale and impact should do the work.
- Programming weak velocities or long MIDI notes at fast tempo.
- Using nonstop fills so there is no pressure-release contrast.
- Mistaking loudness for composition; the loop still needs a distinct rhythmic identity.

Originality mutations for this archetype

- Keep the one-bar rhythm but reduce the hook pitch set from four notes to three.
- Move one root repetition up an octave at the phrase peak.
- Replace bVII motion with a pedal tone and rare bII tension hit.
- Create the second drop by subtraction before adding ride energy.

PROFILE 4

AVICII

AVICII: Melody-first progressive house and emotionally direct harmony

Documented context

Avicii publicly said he started with melody and would sit at the piano to play chords with the melody. Image-Line's artist interview also describes melody as a durable core idea that was refined in harmony and energy. The implementation details below are generalized analysis.

Generator identity

PARAMETER	DEFAULT / RANGE
Tempo	124-128 BPM; default 126
Core meter	4/4
Harmonic loop	4-8 bars; songlike harmonic sentences
Chord density	Medium; usually clear full progression
Melody density	Medium to high in hooks; singable contour
Swing / microtiming	0.00-0.10; generally straight and precise
Kick policy	Stable four-on-floor in dance sections
Primary energy lever	Chord emotion, memorable lead, octave layering, arrangement lift

Listening-based analysis

- The melody is the identity layer. Chords support its emotional reading and should remain clear enough for the hook to be singable.
- Use familiar diatonic functions with strong voice leading rather than complexity for its own sake.
- A lead can outline chord tones while using passing notes to create forward motion. Repetition is balanced by small tail changes and phrase peaks.
- The arrangement creates emotional scale by moving between intimate piano/guitar-like presentations and full dance layers.

Harmonic engine

ID	ROMAN NUMERALS	EXAMPLE IN KEY	GENERATOR USE
A1	I - V - vi - IV	C major: C - G - Am - F	Canonical bright-emotional loop; vary inversion and top voice
A2	vi - IV - I - V	G major: Em - C - G - D	Minor opening with major release
A3	I - iii - IV - vi	D major: D - F#m - G - Bm	Gentler lift; useful under vocal-like melody
A4	i - VI - III - VII	B minor: Bm - G - D - A	Anthemic minor loop with relative-major openness
A5	IV - I - V - vi	A major: D - A - E - F#m	Starts open and delays tonic emphasis
A6	I - V/3 - vi - IV	E major: E - B/D# - C#m - A	Bass-line voice leading; strong songwriting motion

VOICING RULE

Begin with piano-friendly triads and inversions. Keep the top voice smooth and let the bass descend or rise purposefully. Add sus2, add9, or octave doublings for openness, but do not obscure the chord function. In drops, layer the same MIDI across complementary sounds rather than adding unrelated notes.

Melody engine

- Write a 1-2 bar motif that can be sung back after one hearing.
- Anchor important onsets to chord tones, especially thirds and roots.
- Use stepwise passing motion between chord targets.
- Create one clear register peak in bars 3-4 or 7-8.
- Repeat the motif with a changed ending; avoid constant new material.
- Use held notes or rests before a drop so the next attack feels larger.
- Test on plain piano. If the hook needs production to make sense, rewrite it.

One-bar drum skeleton

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Open hat			X				X				X				X	
Shaker		o		o		o		o		o		o		o		o
Build snare													o		o	X

Treat this grid as a skeleton. Primary hits are deterministic; open-dot positions are ghost or probability-controlled. Vary velocity and microtiming within the profile's groove range.

Bass engine

Follow roots clearly, often on offbeat eighths or a sustained/pulsed pattern. Use inversions in the chord progression to create a melodic bass line. Passing notes should connect roots, not compete with the lead.

Arrangement state sequence

STATE	LENGTH	MIDI ACTION
Intro / verse	8-16 bars	Piano, guitar-like layer, or filtered harmony; space for vocal
Pre-lift	8 bars	Motif hint, added pulse, rising register
Chorus / build	8-16 bars	Full harmony and vocal/hook expectation
Drop A	16 bars	Lead states motif over four-on-floor
Verse / break	8-16 bars	Return to song core with reduced dance layers
Drop B	16-32 bars	Hook variation, octave layer, added counterline or ride

Parameter card

```
{
  "archetype": "melody_first_progressive_house",
  "tempo": {
    "min": 124,
    "max": 128,
    "default": 126
  },
  "key_mode_weights": {
    "major": 0.54,
    "minor": 0.46
  },
  "progression_lengthBars": [
    4,
    8
  ],
  "lead_chord_tone_weight_on_strong_beats": 0.78,
  "lead_range_semitones": [
    9,
    16
  ],
  "phrase_peak_location": [
    0.58,
    0.82
  ],
  "swing_range": [
    0.0,
    0.1
  ]
}
```

Generation recipe

- Choose a tempo inside 124-128 BPM and select a key from the weighted mode distribution.
- Select one progression template, transpose it, and generate three inversion candidates. Keep the lowest voice-leading-cost version.
- Generate a role-correct bass rhythm before decorative percussion. Reject collisions that weaken the kick.
- Create a short motif from scale degrees, then fit strong onsets to chord tones. Produce statement, answer, lift, and cadence variants.
- Apply the drum skeleton, optional-event budget, lane velocities, swing, and microtiming.
- Map the phrase into the arrangement state sequence. Every 4-8 bars, alter only one or two attributes.
- Run originality, range, density, collision, and phrase-coherence tests before export.

Failure modes

- Starting from a sound-design trick without a durable melody and chord core.
- Using too many extensions or chromatic chords, clouding emotional directness.
- Writing a lead that never repeats or has no clear phrase peak.
- Copying the interval rhythm of a famous hook and merely transposing it.

Originality mutations for this archetype

- Retain the chord functions but design a new top-line rhythm before assigning pitches.
- Change the phrase peak from bar 4 to bar 3 and rewrite the cadence.
- Convert one repeated note into a stepwise passing pair.
- Use a new inversion bass line and a different opening scale degree.

PROFILE 5

DAVID GUETTA

DAVID GUETTA: Pop-EDM clarity, vocal space, and scalable festival structure

Documented context
David Guetta has publicly emphasized that emotion comes from the music and chord progression. Sound On Sound reporting on his collaborations describes choosing among chord progressions and then developing the song and production. The rules below are broad pop-EDM analysis.

Generator identity

PARAMETER	DEFAULT / RANGE
Tempo	124-130 BPM for classic pop-EDM; preset can extend by era
Core meter	4/4
Harmonic loop	4-8 bars; chord progression supports topline
Chord density	Medium; clear harmonic skeleton
Melody density	Topline-led; instrumental hook complements vocal
Swing / microtiming	0.00-0.12; straight dance precision
Kick policy	Stable four-on-floor in dance sections
Primary energy lever	Songwriting core, vocal hook, build/drop contrast, broad voicings

Listening-based analysis

- The harmonic loop must support a strong vocal or lead topline. Clarity wins over ornamental harmony.
- Separate the vocal-hook register from the instrumental-hook register or alternate them in time.
- Build/drop structure scales a song core: remove bass and drums to expose emotion, then restore width, sub, and rhythmic drive.
- Collaboration-aware generation matters: produce slots for vocal, pre-chorus lift, post-chorus hook, and DJ-functional intros/outros.

Harmonic engine

ID	ROMAN NUMERALS	EXAMPLE IN KEY	GENERATOR USE
G1	vi - IV - I - V	C major: Am - F - C - G	Pop-EDM workhorse; strong for vocal chorus
G2	i - bVII - bVI	C minor: Cm - Bb - Ab	Direct minor drama; three-chord cycle
G3	I - V - IV - IV	D major: D - A - G - G	Open, simple, vocal-friendly
G4	IV - V - vi - vi	E major: A - B - C#m - C#m	Pre-chorus lift and delayed resolution
G5	i - VI - III - VII	F# minor: F#m - D - A - E	Festival minor-to-major openness
G6	I - iii - vi - IV	Bb major: Bb - Dm - Gm - Eb	Smooth pop movement and topline support

VOICING RULE

Use piano or guitar-like triads to validate the song, then expand to octaves, fifth layers, and wide upper stacks. Keep the midrange clear for vocals. Sus2 and add9 colors are safe when they do not conflict with the topline. Drop voicings can be broad but should still share one harmonic skeleton.

Melody engine

- Reserve the strongest motif for either vocal chorus or post-chorus lead; do not make both equally busy.
- Use 3-6 note pitch vocabularies for highly memorable chorus cells.
- Repeat the title or hook rhythm with one lift note near the end.
- Aim for a clear two- or four-bar question-and-answer shape.
- Keep strong notes chord-aware and let passing notes connect them.
- For instrumental drops, simplify the chorus contour into a more rhythmic synth cell rather than duplicating every vocal note.
- Allow one pre-chorus ascent that raises pitch, density, or harmonic tension.

One-bar drum skeleton

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Open hat			X				X				X				X	
Top loop		o		o		o		o		o		o		o		o
Build snare									X		X		X	X	X	X

Treat this grid as a skeleton. Primary hits are deterministic; open-dot positions are ghost or probability-controlled. Vary velocity and microtiming within the profile's groove range.

Bass engine

Use clear roots, fifths, and octave pulses. In vocal sections, simplify or sustain; in drops, use offbeat or eighth-note propulsion. Follow slash-chord inversions when they improve the bass melody, but keep sub movement predictable enough for festival translation.

Arrangement state sequence

STATE	LENGTH	MIDI ACTION
Intro	8-16 bars	DJ-compatible rhythm or immediate song hook
Verse	8-16 bars	Reduced harmony and drums; vocal space
Pre-chorus	8 bars	Rising density, harmonic tension, snare or pulse
Chorus / drop	16 bars	Full bass, kick, wide harmony, hook
Verse 2 / break	8-16 bars	New supporting detail, reduced energy
Final chorus	16-32 bars	Added harmony, octave, counterline, or extended outro

Parameter card

```
{
  "archetype": "pop_edm_clarity",
  "tempo": {
    "min": 124,
    "max": 130,
    "default": 128
  },
  "key_mode_weights": {
    "major": 0.48,
    "minor": 0.52
  },
  "vocal_space_priority": 0.88,
  "chorus_pitch_vocabulary_size": [
    3,
    6
  ],
  "drop_width_layer_count": [
    3,
    6
  ],
  "prechorus_rise_probability": 0.82,
  "swing_range": [
    0.0,
    0.12
  ]
}
```

Generation recipe

- Choose a tempo inside 124-130 BPM for classic pop-EDM and select a key from the weighted mode distribution.
- Select one progression template, transpose it, and generate three inversion candidates. Keep the lowest voice-leading-cost version.
- Generate a role-correct bass rhythm before decorative percussion. Reject collisions that weaken the kick.
- Create a short motif from scale degrees, then fit strong onsets to chord tones. Produce statement, answer, lift, and cadence variants.
- Apply the drum skeleton, optional-event budget, lane velocities, swing, and microtiming.
- Map the phrase into the arrangement state sequence. Every 4-8 bars, alter only one or two attributes.
- Run originality, range, density, collision, and phrase-coherence tests before export.

Failure modes

- Filling the midrange with chords and leads that leave no topline space.
- Treating the build as only a snare roll rather than harmonic and melodic expectation.
- Making the drop unrelated to the song's emotional core.
- Overcomplicating chord function when a simple loop would communicate more directly.

Originality mutations for this archetype

- Rewrite the chorus onset pattern while preserving only the phrase length.
- Move the instrumental hook into the gaps of the vocal rather than doubling it.
- Change the pre-chorus ascent into rhythmic compression instead of pitch ascent.
- Replace one chord with a held pedal and new bass inversion.

PART III

Turn the analysis into a working generator

The next chapters convert the musical descriptions into generation order, state, constraints, mutation, scoring, and export behavior.

6. Generation order and dependency graph

Generation order matters because layers depend on one another. Tempo, meter, key, and archetype define the space. Harmony defines stable pitch targets. Kick and harmonic rhythm shape bass. Bass and drums define groove. Melody negotiates with harmony and vocal space. Arrangement decides when each layer appears. Humanization is applied after the symbolic structure is accepted.

Do not generate all tracks independently and hope they align. Use shared objects: every note can query the current chord, section energy, kick mask, density budget, and reserved register.

Recommended pipeline

STEP	MODULE	OUTPUT	HARD CHECK
1	Global setup	Tempo, key, scale, PPQ, archetype	Valid range and deterministic seed
2	Form	Section states and energy curve	Total length and transition logic
3	Harmony	Roman-numeral loop and voicings	Scale fit and voice-leading cost
4	Kick + backbeat	Anchor masks	Stable floor; no unintended gaps
5	Bass	Pitch and rhythm cell	Kick collision and chord fit
6	Chord rhythm	Stabs, sustains, or gates	Register and bass separation
7	Melody / hook	Motif plus phrase variants	Chord-tone, range, and originality checks
8	Hats + percussion	Groove detail	Density budget and timing coherence
9	Section mutation	Fills, mutes, lifts, reductions	Only controlled deltas per phrase
10	Humanization	Velocity, chance, microtiming	Anchor protection
11	Export	MIDI tracks plus metadata	Note bounds, ordering, no stuck notes

Shared context object

```
{
  "seed": 82041,
  "ppq": 480,
  "tempo_bpm": 127,
  "meter": [
    4,
    4
  ],
  "key": "G",
  "mode": "minor",
  "archetype": "minimal_deep_tech_pocket",
  "section": {
    "name": "groove_A",
    "bar": 17,
    "energy": 0.68
  },
  "current_chord": {
    "degree": 1,
    "quality": "minor7"
  },
  "kick_mask_16": "x...x...x...x...",
  "reserved_registers": {
    "sub": [
      28,
      48
    ],
    "vocal": [
      60,
      79
    ],
    "lead": [
      72,
      91
    ]
  },
  "density_budget": {
    "notes_per_bar": 22,
    "optional_drum_hits": 7
  }
}
```

7. Weighted constraints, not random choice

A useful generator combines hard constraints, weighted preferences, and controlled randomness. Hard constraints prevent invalid or unmusical output: notes outside MIDI range, overlapping monophonic bass events, melody outside its register, or too many optional drum hits. Soft scores decide among valid candidates.

Generate several candidates at each important stage, score them, and sample from the top group rather than always choosing the single maximum. This preserves diversity without accepting obviously weak phrases.

Candidate score

```
score =
  1.6 * harmony_fit
+ 1.4 * groove_fit
+ 1.2 * motif_coherence
+ 1.0 * voice_leading
+ 0.9 * phrase_shape
+ 0.8 * archetype_fit
+ 0.6 * playable_range
+ 0.5 * novelty
- 1.8 * collision_penalty
- 2.2 * reference_similarity
- 1.0 * density_violation
```

Hard versus soft examples

RULE	TYPE	EXAMPLE
Scale membership	Soft	Chromatic approaches allowed with penalty or explicit rule
Strong-beat chord fit	Soft	Target 70-85%, not 100%
MIDI note 0-127	Hard	Reject out-of-range note
Bass monophony	Hard by default	No overlapping sub notes unless legato mode
Melody range	Hard within profile	Reject phrase beyond selected register
Optional drum budget	Hard	Do not exceed per-bar count
Motif repetition	Soft	Reward recognizable reuse but permit answer variation
Reference similarity	Hard threshold	Reject combined pitch-rhythm match

Top-k stochastic selection

After scoring, keep the top 10-25 percent of candidates and sample using a temperature parameter. Lower temperature produces a more consistent preset; higher temperature explores more unusual but valid results. Never use temperature to relax hard originality or MIDI-validity checks.

8. Phrase, section, and arrangement state machines

A loop generator becomes a track generator only when it can create expectation and change. Define arrangement as a state machine. Each state controls active layers, energy, register, note density, harmonic rhythm, and transition probability. The archetype provides the most likely state order, but the exact bar counts can vary within allowed ranges.

Transitions should produce musical deltas. A lift may remove the kick, lengthen chords, raise melody register, and reduce sub energy. A drop restores kick and bass, shortens chord articulation, and states the main motif. The model should know which changes cause energy, rather than merely adding tracks.

Generic house state model

STATE	ENERGY	ACTIVE CORE	TRANSITION LEVER
INTRO	0.20-0.40	Kick or filtered drums, texture	Add bass or defining percussion
GROOVE_A	0.50-0.70	Kick, bass, core drums, restrained hook	Expose harmony or remove low end
LIFT	0.35-0.65	Harmony, vocal/lead hint, reduced floor	Riser, density ramp, held note
PEAK_A	0.75-0.95	Full groove and primary hook	Ride, octave, wider chord
BREAK	0.20-0.50	Song core or sample, minimal drums	Rebuild pulse and anticipation
PEAK_B	0.82-1.00	Full return plus one meaningful variation	Counterline, new tail, longer final phrase
OUTRO	0.20-0.55	DJ-functional drums, reduced hooks	Subtract bass and signature content

Energy control vector

```
{
  "energy": 0.84,
  "layer_count": 10,
  "kick_strength": 1.0,
  "sub_presence": 0.95,
  "top_end_density": 0.78,
  "chord_register": 0.72,
  "melody_register": 0.81,
  "note_density": 0.68,
  "stereo_width_hint": 0.82,
  "transition_fx_hint": 0.35
}
```

Variation budget

Within an established section, mutate one or two salient properties every 4-8 bars: remove a kick, add a pickup, change the final bass note, open a hat, extend the chord tail, or answer the motif. At a major section boundary, change three or more coordinated properties. This keeps micro-variation from competing with macro-form.

9. Originality and similarity control

Artist-associated presets increase the risk that a generator will lean on a known record. Protect originality explicitly. The best safeguard is not a disclaimer; it is a measurable rejection layer that compares symbolic fingerprints and requires independent structural mutations.

Normalize melodies before comparison by converting absolute pitches to intervals, onsets to beat-relative gaps, and durations to ratios. Compare short n-grams as well as whole phrases. A transposed copy will retain its interval and rhythm fingerprints and should still be detected.

Similarity features

FEATURE	NORMALIZATION	USE
Pitch interval sequence	Subtract consecutive MIDI pitches	Detect transposed melodic copying
Rhythm interval sequence	Divide onset gaps by beat length	Detect same rhythm at new tempo
Duration ratio	Normalize to local subdivision	Detect articulation identity
Contour	Map intervals to up / same / down	Coarse shape similarity
Strong-beat degrees	Convert to scale degree over chord	Detect harmonic placement
Chord-function sequence	Roman numerals	Avoid entire progression-plus-rhythm identity
Drum onset masks	16-step binary lanes	Avoid copying signature drum breaks

Rejection policy

```

if melody_interval_similarity > 0.86
and melody_rhythm_similarity > 0.82:
    reject("hook too close")

if drum_mask_similarity > 0.90
and fill_location_match:
    mutate_drums()

if chord_sequence_match
and harmonic_rhythm_match
and bass_rhythm_similarity > 0.84:
    rewrite_two_independent_layers()

```

Minimum independent mutation set

- Change the onset pattern, not only the pitches.
- Change at least one phrase boundary or cadence behavior.
- Change the contour peak location.
- Change voicing and bass motion independently.
- Change drum or chord articulation if the reference identity depends on rhythm.
- Re-run similarity after mutation; do not assume changes are sufficient.

10. End-to-end pseudocode

```
function generate_track(request, seed):
    rng = seeded_random(seed)
    preset = load_archetype(request.archetype)
    global = choose_global_settings(request, preset, rng)
    form = build_arrangement_states(preset, request.length, rng)

    progression = sample_progression(preset, global.key, rng)
    voiced_chords = optimize_inversions(progression, preset.voicing)
    assert harmony_is_valid(voiced_chords, global.scale)

    kick = generate_kick(form, preset.kick_policy, rng)
    clap = generate_backbeat(form, preset, rng)
    bass_candidates = generate_bass(voiced_chords, kick, preset, rng, count=32)
    bass = select_top_k(bass_candidates, score_bass, temperature=0.35)

    chord_rhythm = articulate_chords(voiced_chords, form, preset, rng)
    motif_seed = generate_motif(progression, preset.melody, rng)
    phrase_candidates = develop_phrase(motif_seed, progression, preset, count=48)
    phrase = select_top_k(phrase_candidates, score_phrase, temperature=0.28)

    hats = generate_hats(kick, clap, preset.groove, rng)
    percussion = generate_optional_percussion(
        anchors=[kick, clap, hats],
        budget=preset.optional_drum_density_budget,
        rng=rng
    )

    tracks = map_layers_to_form(
        form, kick, clap, hats, percussion, bass, chord_rhythm, phrase
    )
    tracks = add_transition_events(tracks, form, preset, rng)
    tracks = apply_velocity_and_microtiming(tracks, preset, rng)

    assert midi_bounds_valid(tracks)
    assert density_valid(tracks, preset)
    assert groove_coherent(tracks, preset)
    assert reference_similarity(tracks) < request.similarity_threshold

    return export_midi(tracks, metadata={global, form, seed, preset.version})
```

Determinism and seeds

Every generation should save its seed, preset version, global settings, and mutation history. The same seed and code version should recreate the same MIDI. This makes debugging possible and lets a user lock one layer while regenerating another.

Layer locking

```
{
  "lock": [
    "tempo",
    "key",
    "chords",
    "kick"
  ],
  "regenerate": [
    "bass",
    "melody",
    "hats"
  ],
  "preserve_phrase_boundaries": true,
  "mutation_strength": 0.38
}
```

11. Validation suite

Automated checks should be paired with listening, but they can eliminate many broken outputs. Run tests at event, bar, phrase, section, and track levels. A generator brain becomes reliable when failures are named and measurable.

Event-level tests

- Pitch is an integer from 0 to 127.
- Velocity is an integer from 1 to 127 for note-on events.
- Duration is positive; paired note-off occurs after note-on.
- Events are sorted by performed time, then track priority.
- Monophonic bass notes do not overlap unless legato mode is enabled.

Bar-level tests

- Kick anchors match the selected policy.
- Optional drum hits do not exceed the density budget.
- Chord and bass pitches agree with the active harmony or an explicit approach rule.
- No register has excessive simultaneous note count.
- Microtiming does not reorder important anchors.

Phrase-level tests

- Motif appears in recognizable form at least twice.
- Statement and answer differ meaningfully but share an identity.
- Phrase has one main peak or deliberate flat/hypnotic profile.
- Cadence either closes or creates a clear pickup into the loop.
- Interval and rhythm fingerprints pass originality thresholds.

Section-level tests

- Energy direction matches the arrangement state.
- A transition changes at least the minimum number of coordinated features.
- The main hook is not active at full intensity in every section.
- Bass and kick return together only when intended.
- Fills occur near boundaries, not randomly throughout the section.

Listening checklist

SOLO / MUTE TEST	QUESTION	IF NO
Piano only	Do chords and melody communicate an idea?	Rewrite harmony or motif before sound design
Kick + bass	Does the low-end rhythm feel intentional?	Change onset relation and note lengths
Kick + clap + hat	Is the groove readable?	Simplify and repair hierarchy
No drums	Does the track still have phrase direction?	Improve harmonic/melodic form
No melody	Does the groove sustain interest?	Strengthen bass and micro-variation
Mono playback	Do essential roles remain clear?	Reduce register collisions and excess layering

PART IV

Libraries, schemas, and quick-reference appendices

Appendix A. Progression library

These are function templates, not artist transcriptions. Transpose freely, change harmonic rhythm, and use archetype-specific voicing. The mood labels are tendencies rather than guarantees.

ID	FUNCTIONS	CHARACTER	MODE	LOOP
L01	I - V - vi - IV	Bright, emotional, direct	Major	4 bars
L02	vi - IV - I - V	Minor opening, major release	Major / relative minor	4 bars
L03	i - VI - III - VII	Anthemic minor openness	Natural minor	4 bars
L04	i - bVII - bVI - bVII	Rave descent and return	Natural minor	4 bars
L05	I - bVII	Blunt, open, Mixolydian	Mixolydian	2 bars
L06	i - VI	Pressure, simplicity	Minor	2 bars
L07	i7 - iv7	Deep, circular	Minor	2-4 bars
L08	ii7 - V7	Jazz tension without tonic	Major	2 bars
L09	i9 - bVIImaj7	Warm deep-house color	Minor	2 bars
L10	I6/9 pedal	Static, hypnotic	Major	2-8 bars
L11	IV - V - vi	Rising pre-chorus	Major	2-4 bars
L12	I - V/3 - vi - IV	Smooth bass voice leading	Major	4 bars
L13	IV - I - V - vi	Open start, delayed tonic	Major	4 bars
L14	i - iv - VI - V	Dramatic minor cadence	Harmonic minor option	4 bars
L15	I - iii - IV - vi	Gentle progressive lift	Major	4 bars
L16	vi7 - ii7 - V7	Longer deep-house motion	Major	3-6 bars
L17	i pedal + bII accent	Dark tension	Phrygian color	1-4 bars
L18	I - IV - ii - V	Classic forward movement	Major	4 bars
L19	i - III - VII - iv	Dark with modal breadth	Minor	4 bars
L20	I - IV - vi - V	Familiar with softer second half	Major	4 bars
L21	vi - V - IV - V	Looping longing	Major / relative minor	4 bars
L22	i - v - VI - iv	Brooding and circular	Minor	4 bars
L23	I - ii7 - IV - I	Warm, gospel-adjacent color	Major	4 bars
L24	i - bVII - iv - VI	Underground tension	Minor	4 bars

Transposition example

Template L02 is vi - IV - I - V. In C major it is Am - F - C - G. In E major it becomes C#m - A - E - B. The Roman numerals and scale-degree melody stay the same; absolute MIDI pitches change. Recalculate inversions after transposition rather than shifting every note blindly if the new register becomes too high or low.

Appendix B. Drum pattern library

Straight pop-house

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Open hat			X				X				X				X	
Shaker		o		o		o		o		o		o		o		o

Shuffled minimal pocket

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Closed hat			X				X				X				X	
Shaker		X		X		X		X		X		X		X		X
Rim				o			X					o			X	
Perc		o				X		o		o				X		

Rave-house hybrid

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	X				X				X				X			
Clap					X								X			
Break ghost				o			X					o			X	
Closed hat			X		X		X				X		X		X	
Ride	o		o		o		o		o		o		o		o	
Fill													o		o	X

Hard-loop pressure

LANE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Kick	X				X				X				X				
Clap					X								X				
Closed hat			X		X		X				X		X		X		
Open hat			X				X				X				X		
Ride	X		X		X		X		X		X		X		X		
Fill				o			X					o		X	X	X	

Pattern mutation rules

- Rotate only optional percussion, never the full anchor pattern.
- For a fill, replace the last one or two beats rather than adding on top of all existing hits.
- If hat density increases, reduce shaker or ride probability to respect a top-end budget.
- Change velocity and microtiming before adding new onsets.
- Use a two-bar pattern when a one-bar loop feels too predictable; change bar 2 with one or two events.

Appendix C. Archetype comparison matrix

ARCHETYPE	TEMPO	CHORDS	MELODY	GROOVE	MAIN ENERGY LEVER
PROSPA	124-134 BPM	Medium to high in peaks	Motif-led, medium	0.06-0.20	Piano/organ stabs, rave motifs, breaks, octave lift
CHRIS STUSSY	124-130 BPM	Low	Low to medium	0.14-0.30	Bass rhythm, shuffle, subtractive arrangement, micro-variation
KETTAMA	130-145 BPM	Low to medium	Low to medium	0.04-0.18	Drum impact, sample repetition, chord scale, tension drops
AVICII	124-128 BPM	Medium	Medium to high in hooks	0.00-0.10	Chord emotion, memorable lead, octave layering, arrangement lift
DAVID GUETTA	124-130 BPM for classic pop-EDM	Medium	Topline-led	0.00-0.12	Songwriting core, vocal hook, build/drop contrast, broad voicings

Fast preset selection

- **Choose Prospa-associated rave-house** when the request emphasizes piano/organ stabs, rave lift, and breakbeat memory.
- **Choose Chris Stussy-associated minimal/deep tech** when bass rhythm, swing, restraint, and long groove development are primary.
- **Choose KETTAMA-associated high-impact pressure** when the request asks for fast, hard, sample-led club force.
- **Choose Avicii-associated melodic progressive house** when a memorable instrumental melody and emotionally direct chords lead.
- **Choose David Guetta-associated pop-EDM** when a vocal-ready song core, build/drop clarity, and broad accessibility lead.

Hybridization policy

Hybrid presets should name one primary archetype and at most two borrowed dimensions. Example: primary Stussy pocket, borrowed Prospa chord articulation, and Avicii phrase clarity. Do not average every parameter; that produces an indistinct center. Preserve the primary preset's tempo, density, and arrangement, then borrow a specific melody, harmony, or drum behavior.

```
{
  "primary": "minimal_deep_tech_pocket",
  "borrow": [
    {
      "from": "rave_house_lift",
      "dimension": "chord_articulation",
      "amount": 0.35
    },
    {
      "from": "melody_first_progressive_house",
      "dimension": "phrase_shape",
      "amount": 0.2
    }
  ],
  "locked_primary_dimensions": [
    "tempo",
    "swing",
    "bass_density",
    "arrangement"
  ]
}
```

Appendix D. Prompt templates for a MIDI generator

Full-track prompt

Generate an original 128 BPM house MIDI arrangement in A minor.
Primary archetype: rave-house lift.
Use a new 4-bar progression selected from functional templates; do not imitate any released hook. Chords should be short piano/organ stabs with smooth top-voice movement. Create a 2-bar lead motif with 3-5 distinct pitches, one pickup, and one octave accent. Use four-on-the-floor kick with a restrained breakbeat ghost layer. Create a root-and-octave bass pattern that leaves kick space. Arrange 96 bars: intro, pressure A, lift, peak A, switch, peak B, outro.
Return separate MIDI tracks plus JSON metadata and similarity-test results.

Regenerate only the melody

Keep tempo, key, chords, bass, drums, and section boundaries locked.
Generate 24 melody candidates for the selected archetype. Strong beats should favor chord tones, total range must be 9-14 semitones, and the phrase needs one peak between 60% and 80% of its duration. Select from the top 20% by motif coherence and originality. Reject any candidate whose interval and rhythm fingerprints both exceed the reference thresholds.

Generate a two-bar minimal groove

Create a 2-bar 127 BPM deep-tech MIDI groove in G minor.
Kick: stable four-on-floor. Clap: beats 2 and 4. Bass: 5-8 notes across two bars, mostly root/fifth/octave plus at most one chromatic approach. Apply 0.18 swing to eligible sixteenths. Add hats and percussion under an optional-hit budget of 7 per bar. Chord: one compact Gm9/Cm7 shell gesture with shared tones. Make bar 2 differ by only one bass event and one percussion event.

Machine-output contract

```
{
  "midi_files": [
    "kick.mid",
    "clap.mid",
    "hats.mid",
    "percussion.mid",
    "bass.mid",
    "chords.mid",
    "lead.mid"
  ],
  "metadata_required": [
    "seed",
    "tempo_bpm",
    "ppq",
    "key",
    "mode",
    "archetype",
    "progression_roman",
    "section_map",
    "similarity_scores"
  ],
  "validation_required": true
}
```

Appendix E. Compact JSON schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "HouseMidiGeneratorPreset",
  "type": "object",
  "required": [
    "id",
    "tempo",
    "harmony",
    "melody",
    "drums",
    "bass",
    "arrangement"
  ],
  "properties": {
    "id": {
      "type": "string"
    },
    "tempo": {
      "type": "object",
      "required": [
        "min",
        "max",
        "default"
      ],
      "properties": {
        "min": {
          "type": "number"
        },
        "max": {
          "type": "number"
        },
        "default": {
          "type": "number"
        }
      }
    },
    "harmony": {
      "type": "object",
      "properties": {
        "progression_ids": {
          "type": "array",
          "items": {
            "type": "string"
          }
        },
        "extension_probabilities": {
          "type": "object"
        },
        "harmonic_rhythm_steps": {
          "type": "array",
          "items": {
            "type": "integer"
          }
        },
        "voicing_register": {
          "type": "array",
          "minItems": 2,
          "maxItems": 2
        }
      }
    },
    "melody": {
      "type": "object",
      "properties": {
        "distinct_pitch_range": {
          "type": "array"
        },
        "total_range_semitones": {
          "type": "array"
        },
        "strong_beat_chord_tone_weight": {
          "type": "number"
        },
        "mutation_operators": {
          "type": "array"
        }
      }
    },
    "drums": {
      "type": "object",
      "properties": {
        "kick_mask": {
```

```

        "type": "string",
        "pattern": "^[xo.]{16}$"
    },
    "clap_mask": {
        "type": "string",
        "pattern": "^[xo.]{16}$"
    },
    "swing_range": {
        "type": "array"
    },
    "optional_hit_budget": {
        "type": "integer"
    }
}
},
"bass": {
    "type": "object",
    "properties": {
        "pitch_weights": {
            "type": "object"
        },
        "notes_per_bar": {
            "type": "array"
        },
        "monophonic": {
            "type": "boolean"
        }
    }
},
"arrangement": {
    "type": "array",
    "items": {
        "type": "object"
    }
}
}
}

```

Chunking advice for retrieval systems

- Chunk by heading, not fixed character count, whenever possible.
- Keep an archetype's parameter card with its explanation.
- Store table captions and row text together.
- Attach metadata: archetype, layer, musical topic, evidence label, and page number.
- Retrieve one primary archetype section plus only the shared theory sections needed for the requested layer.
- Do not retrieve every artist profile for a single generation unless the request explicitly asks for a hybrid.

Appendix F. Glossary

TERM	DEFINITION
Anchor	A structurally reliable event such as a kick on each quarter note or clap on beats 2 and 4.
Archetype	A parameter preset capturing broad musical behaviors; not an artist clone.
Chord tone	A pitch belonging to the current chord, usually root, third, fifth, seventh, or chosen extension.
Contour	The up, down, and repeated-note shape of a melody independent of exact key.
Density budget	A maximum amount of note or onset activity allowed in a bar or layer group.
Ghost note	A low-velocity or probability-controlled event that supports groove without becoming an anchor.
Harmonic rhythm	The rate and placement at which chords change or articulate.
Inversion	A chord voicing with a chord tone other than the root in the lowest position.
Microtiming	Small performance offsets around the quantized grid.
Motif	A short, recognizable pitch-rhythm idea developed across a phrase.
Onset mask	A step-grid representation showing where events begin.
PPQ	Pulses per quarter note; MIDI time resolution.
Roman numeral	A key-relative label for chord function, enabling transposition.
Shell voicing	A reduced chord voicing, often third and seventh or root and seventh.
Strong beat	A metrically important location, especially beat 1 and other quarter-note anchors.
Swing	Systematic delay of selected subdivisions to create a long-short feel.
Top voice	The highest note in a chord voicing, often perceived melodically.
Voice leading	The movement of individual chord voices from one chord to the next.

References and source notes

Accessed 23 July 2026. These sources support documented workflow context and general MIDI implementation concepts. All prescriptive parameter ranges, progression banks, pattern masks, and pseudocode in this document are original synthesis and analysis.

#	SOURCE	RELEVANCE	URL
1	Avicii - Image-Line artist interview	Avicii discusses FL Studio, melody as an early core idea, and refinement of harmony and energy.	https://www.image-line.com/artists/avicii-
2	Veni, Vidi, Avicii - Interview Magazine	Avicii states that he starts with the melody and sits at the piano with chords and melody.	https://www.interviewmagazine.com/music/avicii-dj
3	In pictures: Avicii's Stockholm studio - MusicRadar	Studio and workflow context, including Avicii's use of FL Studio.	https://www.musicradar.com/news/tech/in-pictures-aviciis-stockholm-studio-508635
4	Artist Interview: Chris Stussy - Sample Market	Public comments on studio discipline and drum-machine use inside Fruity Loops / FL Studio.	https://samplemarket.co.uk/blog/artist-interview-chris-stussy
5	Chris Stussy: house music all night long - DJ Mag	Career and production context, including continued FL Studio use.	https://djmag.com/features/chris-stussy-house-music-all-night-long-interview
6	Introducing: KETTAMA - Beatportal	KETTAMA describes an intuitive, sample-heavy FL Studio process and loop-led club music.	https://www.beatportal.com/articles/10950-introducing-kettama
7	KETTAMA: The DJ Of The Moment - Mixmag	Production background and discussion of large chords, basslines, and forceful drums.	https://mixmag.net/feature/kettama-cover-feature-interview-archangel-dj
8	David Guetta - Extended Interview - Musician's Friend	Guetta emphasizes emotion, music, and chord progression in his production process.	https://www.musiciansfriend.com/david-guetta-extended-interview
9	David Guetta - Sound On Sound	Collaboration and song-development context, including choosing a chord progression.	https://www.soundonsound.com/people/david-guetta
10	Editing MIDI - Ableton Live 12 Manual	Official documentation for per-note MIDI editing, velocity, probability, and related controls.	https://www.ableton.com/en/live-manual/12/editing-midi/
11	Using Grooves - Ableton Live 12 Manual	Official documentation for groove timing, randomization, velocity, extraction, and committing grooves.	https://www.ableton.com/en/live-manual/12/using-grooves/
12	Piano roll - FL Studio Manual	Official documentation for piano-roll notes, chords, scales, and note properties.	https://www.image-line.com/fl-studio-learning/fl-studio-online-manual/html/pianoroll.htm

Final implementation principle

Build a musician, not a slot machine

The generator should know why a note exists: harmony target, motif identity, groove role, energy transition, or expressive variation. Randomness should choose among musically valid options. It should never be the substitute for structure.

End of document. Version 1.0. Artist names are used solely to organize independent analysis of broad musical traits. No affiliation or endorsement is implied.